

EXAMPLE 6.6

Consider the following arithmetic infix expression Q:

$$Q: A + (B * C - (D / E \uparrow F) * G) * H$$

We simulate Algorithm 6.4 to transform Q into its equivalent postfix expression P.

First we push "(" onto STACK, and then we add ")" to the end of Q to obtain:

Q: A + (B * C - (D / E \uparrow F) * G) * H)
 (1) (2) (3) (4) (5) (6) (7) (8) (9) (10) (11) (12) (13) (14) (15) (16) (17) (18) (19) (20)

The elements of Q have now been labeled from left to right for easy reference. Figure 6-8 shows the status of STACK and of the string P as each element of Q is scanned. Observe that

- (1) Each operand is simply added to P and does not change STACK.
- (2) The subtraction operator (-) in row 7 sends * from STACK to P before it (-) is pushed onto STACK.
- (3) The right parenthesis in row 14 sends \uparrow and then / from STACK to P, and then removes the left parenthesis from the top of STACK.
- (4) The right parenthesis in row 20 sends * and then + from STACK to P, and then removes the left parenthesis from the top of STACK.

After Step 20 is executed, the STACK is empty and

P: A B C * D E F \uparrow / G * - H * +

which is the required postfix equivalent of Q.

Symbol Scanned	STACK	Expression P
(1) A	(	A
(2) +	(+	A
(3) (	(+ (	A
(4) B	(+ (	A B
(5) *	(+ (*	A B
(6) C	(+ (*	A B C
(7) -	(+ (-	A B C *
(8) (	(+ (- (	A B C *
(9) D	(+ (- (	A B C * D
(10) /	(+ (- (/	A B C * D
(11) E	(+ (- (/	A B C * D E
(12) \uparrow	(+ (- (/ \uparrow	A B C * D E
(13) F	(+ (- (/ \uparrow	A B C * D E F
(14))	(+ (-	A B C * D E F \uparrow /
(15) *	(+ (- *	A B C * D E F \uparrow /
(16) G	(+ (- *	A B C * D E F \uparrow / G
(17))	(+	A B C * D E F \uparrow / G * -
(18) *	(+ *	A B C * D E F \uparrow / G * -
(19) H	(+ *	A B C * D E F \uparrow / G * - H
(20))		A B C * D E F \uparrow / G * - H * +

Fig. 6-8